Random Access Performance for Three Coverage Enhancement Groups in NB-IoT Networks

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***Abstract*—NarrowBand-Internet of Things (NB-IoT) is a new 3GPP radio access technology designed to provide better coverage for Low Power Wide Area (LPWA) networks. To provide reliable connections with extended coverage, a repetition transmission scheme and up to three Coverage Enhancement (CE) groups are introduced into NB-IoT during both Random Access CHannel (RACH) procedure and data transmission procedure, where each CE group is configured with different repetition values. Rather than our previous work only modeled RACH success probability in NB-IoT networks with a single CE group, this paper develops a novel model to analyze the RACH success probabilities in NB-IoT networks with three CE groups, which allow flexible RACH configuration for each CE group. Based on this analytical model, we derive the expression for the RACH success probability of a randomly chosen IoT device in each CE group. The analytical results can also be extended to analyze multiple group users of other networks with spatial separations.**

I. Introduction

There exist several challenges in cellular-based IoT networks, including low device cost, limited latency, massive number of devices, long battery life, and enhanced coverage [1]. To cope with these challenges, the Third Generation Partnership Project (3GPP) has standardized the NB-IoT in Release 13, which defines narrow bandwidth, repetition transmission, single-tone transmission, enhanced discontinuous reception, power spectral density (PSD) boosting, and other network architectural updates [1].

Coverage Enhancement (CE) is one feature for NB-IoT networks, which can be achieved with the help of the narrower carrier bandwidth and the repetition transmission [1]. More importantly, to support various traffic with different coverage conditions, each Base Station (BS) categorizes its IoT devices into up to three CE groups, which provides efficient management of massive number of IoT devices depending on their received signal quality. The RACH repetition value is determined by the BS based on the CE group of the IoT device through the RACH procedure [1].

The main purpose of RACH is to achieve uplink synchronization and obtain the grant for access to the network, in which the first step is to transmit a preamble. Notably, massive connections in NB-IoT may bring simultaneous RACH requests under limited number of available preambles. Thus, it is of great importance to model and analyze the RACH performance of NB-IoT networks, which can be useful for system design and optimization. Importantly, previous results in LTE systems cannot be directly applied to NB-IoT due to its unique characteristics, including transmission repetition, three CE groups configuration, frequency hopping, and etc. Our previous work [2] has provided a general framework to characterize the RACH success probability in NB-IoT networks with repetition scheme based on the preamble transmission model in [3] and collision model in [4]. However, work [2] only considered NB-IoT networks with a single CE group with the transmit power of the IoT device determined by the path-loss inversion power control due to the analytical simplicity, which does not align with the practical NB-IoT networks with multiple CE groups setting. According to the 3GPP standard [5], for the IoT device with repetition value larger than two, its transmission power should be the cell specific maximum transmission power.

Different from [2], we evaluate the RACH success probability taking into account the three geographically separated CE groups in each cell in NB-IoT networks, one challenge of which is how to characterize the number of the active IoT devices using the same preamble (i.e., the interfering IoT devices) in each CE group of the typical cell. The other challenge is how to evaluate the mutual intra-cell interference in each CE group of the typical cell. Generally speaking, in the NB-IoT network with three CE groups, the physical layer parameters and network topology can strongly affect the RACH performance of each CE group.

Even though stochastic geometry has been regarded as a powerful tool to model and analyze mutual interference between transceivers in the wireless networks [6], there are two aspects that limit the application of conventional stochastic geometry analysis to the RACH performance analysis in NB-IoT networks with three CE groups: 1) conventional stochastic geometry works focused on analyzing normal data transmission channel, where the intra-cell interference is not considered, due to the ideal assumption that each orthogonal sub-channel is not reused in a cell, which is not the case when massive IoT devices in each CE group of a cell may randomly choose and transmit the same preamble using the same sub-channel, bringing intra-cell interference; 2) the interference field in conventional stochastic geometry works is mostly modeled by a Poisson Point Process (PPP) to maintain tractability, which is not the case for the interference field in each CE group with spatial separation into three coverage areas among three CE groups.

In this paper, we develop a novel mathematical framework

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for NB-IoT networks with three CE groups. The contributions of this paper can be summarized as follows:

1) We present an analytical framework for the NB-IoT network with three CE groups to analyze the RACH success probability of IoT devices in each CE group.

2) We focus on the approximate characterization of interference in each CE group. We approximate the densities of three CE groups in holes, annulus, and exclusion zones using the Poisson Hole Processes (PHPs) with different densities.

3) We derive the exact expression for the RACH success probability of a randomly chosen IoT device in each CE group taking into account both the preamble transmission outage and the collision.

4) We develop a realistic simulation framework to capture the randomness locations, preamble tranmission as well as the RACH collision, and verify our derived RACH success probability of the IoT device in each CE group.

The rest of the paper is organized as follows. Section II presents a system model. Sections III derives the RACH success probability of a randomly chosen IoT device in each CE group. Our results and simulations are described in Section V. Finally, Section VI has drawn the conclusion.

II. System Model

A. Network Description

We consider a NB-IoT network with three CE groups, where the BSs and IoT devices are spatially distributed following two independent PPPs Φ_{B} and Φ_{D} with intensities λ_{B} and λ_{D}, respectively. We assume each IoT device associates to its nearest BS, where a Voronoi tesselation is formed. We consider additive noise with average power σ^2 and a Rayleigh fading with the channel power gain h assumed to be exponentially distributed with unit mean, i.e., $h \sim$ Exp(1). All channel gains are assumed to be independent and identically distributed (i.i.d.). A standard power-law path-loss model is considered in CE group 0 (described later in Section II.D), where the path-loss attenuation is $x^{-\alpha}$, with the propagation distance x and the path-loss exponent α.

B. Random Access Procedure

The contention-based RACH procedure consists of four steps, where a randomly selected preamble is transmitted to the associated BS on NB-IoT Physical Random Access CHannel (NPRACH), for a given number of times (i.e., the dedicated repetition value) in step 1, and control information with the BS is exchanged in step 2,3,4 [7]. The RACH requests from massive connections in NB-IoT simultaneously under limited number of available preambles is one of the main challenges, thus we focus on the contention of preamble in step 1 of contention-based RACH with the assumption that the step 2,3,4 of RACH are always successful whenever the step 1 is successful following [4]. That is to say a RACH is always successful if the IoT device successfully transmits the preamble to its associated BS. In this case, the RACH success is determined by two reasons: 1) the preamble being successfully transmitted to the associated BS (i.e., received SINR is greater than the SINR threshold γ_{th}); and 2) no collision occurs (i.e., no other IoT devices successfully transmit the same preamble to the typical BS simultaneously). It is known that collision in step 1 can be detected by the BS, when the collided IoT devices are separable in terms of the power delay profile [7]. Our model follows the assumption of collision handling in [4], where collisions are detected by the BS after it decodes the preambles in step 1; hence, the BS will not send the response and the IoT device can not proceed to the next step of RACH.

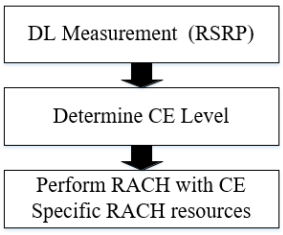


Fig. 1: NB-IoT RACH Procedure

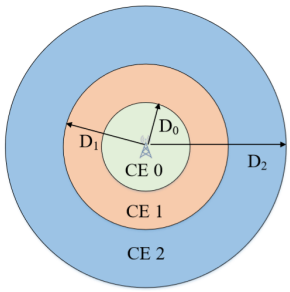


Fig. 2: NB-IoT CE Groups in a single cell

C. Physical Random Access CHannel

In the NPRACH, a preamble consists of four symbol groups transmitted without gaps on a single subcarrier and can be repeated several times using the same transmission power. The subcarrier spacing of NPRACH is 3.75 kHz and up to 48 subcarriers can be allocated to NPRACH. These subcarriers are exclusively shared by three CE groups with a basic subcarrier allocation unit of 12 subcarriers [8]. According to [1], frequency hopping is applied on symbol group granularity, where the first preamble symbol group is transmitted via a subcarrier selected via the pseudo-random hopping (i.e., the hopping depends on the current repetition time and the Narrowband physical Cell ID, a.k.a NCellID [1]), and the following three preamble symbol groups are transmitted via subcarriers determined by the fixed size frequency hopping (i.e. each symbol group is transmitted on a different subcarrier) [8]. That is to say, if two or more IoT devices choose the same first subcarrier, the following subcarriers would be same, due to that these two hopping algorithms lead to one-to-one correspondences between the first subcarrier and the following subcarriers (i.e., these IoT devices either collide on the full set or not collide at all in a single RACH opportunity).

D. CE Group Determination

As shown in Fig. 1, the IoT device determines its CE group by measuring the downlink reference signal received power (RSRP). In this paper, we use the SNR thresholds following [9]. We describe the coverage area, the density, the repetition value and the uplink transmit power in each CE group.

1) Coverage Area of Each CE Group: According to [10], the BS uses the constant power P_{DL} to broadcast the Downlink Control Information (DCI) signal to all the IoT devices

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in its own cell. Based on the received SNRs of DCI signal measured at each IoT device and the SNR thresholds $\{\delta_1, \delta_2\}$, each IoT device independently determine its associated CE group following the rule below:

$$\begin{cases} \dfrac{P_{\mathrm{DL}}x^{-\alpha}}{\omega} \geq \delta_1, & \text{device belongs to CE group 0,} \\ \delta_2 \leq \dfrac{P_{\mathrm{DL}}x^{-\alpha}}{\omega} < \delta_1, & \text{device belongs to CE group 1,} \\ \dfrac{P_{\mathrm{DL}}x^{-\alpha}}{\omega} < \delta_2, & \text{device belongs to CE group 2,} \end{cases} \quad (1)$$

where ω is the noise power at the IoT device and x is the IoT device's distance from the BS.

Thus, the maximum distance D_i between the BS and an IoT device in CE group i can be derived from (1) as

$$\begin{cases} D_0 = \left(\dfrac{\delta_1\omega}{P_{DL}}\right)^{-(1/\alpha)}, \\ D_1 = \left(\dfrac{\delta_2\omega}{P_{DL}}\right)^{-(1/\alpha)}. \end{cases} \quad (2)$$

We define Coverage Area (CA_i) as the area where the N_i IoT devices of CE group i are located in. As shown in Fig. 2, CA_0 is represented by a circle centered at the BS with radius D_0; CA_1 is represented by a annulus centered at the BS with internal radius D_0 and external radius D_1; CA_2 is represented by a annulus centered at the BS with internal radius D_1 and external radius D_2 with D_0, D_1 given in (2) and D_2 given following [11] as

$$D_2 = 1/\sqrt{\pi\lambda_B}. \quad (3)$$

As a consequence, an IoT device belongs to CE group i if it is located in the coverage area CA_i.

2) Density of IoT Devices in Each CE Group: We approximate the interference field Φ_i of each CE group in the following **Lemma** [12].

Lemma 1. *The densities of the interference field Φ_i in the ith CE group λ_i are given as*

$$\begin{cases} \lambda_0 = \lambda_D(1 - \exp(-\lambda_B\pi D_0^2)), \\ \lambda_1 = \lambda_D(\exp(-\lambda_B\pi D_0^2) - \exp(-\lambda_B\pi D_1^2)), \\ \lambda_2 = \lambda_D \exp(-\lambda_B\pi D_1^2), \end{cases} \quad (4)$$

where D_0, D_1 are given in (2) *and D_2 is given in* (3).

The RACH procedure has been modeled as a modified slotted ALOHA, where N_i number of IoT devices contend over a set of time resource assigned by BS. Let S_i denote the number of non-overlapping subcarriers reserved by the BS for CE group i (i.e., $S_0 + S_1 + S_2 \leq 48$). Thus, each preamble in CE group i has an equal probability $(1/S_i)$ to be chosen. The density of IoT devices in CE group i choosing the same preamble can be expressed according to the thinning process as [13]

$$\lambda_i^a = \lambda_i/S_i, \quad (5)$$

where λ_i is given in **Lemma 1**.

3) Repetition Value of IoT devices in Each CE Group: To serve IoT devices in three CE groups, the NB-IoT network can configure three NPRACH resource configurations for each CE group in a cell separately. The BS will notify the NPRACH configuration to the IoT device in the system information by broadcasting, which include the repetition value required for the CE group as well as the preamble transmission power. According to the 3GPP standard [10], the repetition value of the higher CE group is usually larger than that of lower CE group to allow fairness performance among three CE groups, and the repetition value specified for each configuration can be chosen from the sets $R_0 \in \{1, 2\}$ and $R_1, R_2 \in \{4, 8, 16, 32, 64, 128\}$.

4) Uplink Transmit Power of IoT devices in Each CE Group: Based on the 3GPP standard [8], the transmit power in CE group 0 is determined by the path-loss inversion power control, where each IoT device compensates for its own path-loss to keep the average received signal power equal to the same threshold ρ; the transmit power in CE group 1 and 2 is set as a fixed power P (the cell specific fixed transmit power). Therefore, the transmit power of an IoT device in the CE group i can be expressed as

$$\begin{cases} P_{0,k} = \rho(r_{0,k})^{\alpha}, & i = 0, \\ P_{i,k} = P, & i = 1, 2, \end{cases} \quad (6)$$

where $r_{0,k}$ is the distance from the kth IoT device in CE group 0 to the typical BS.

III. Analytical Model

In this section, we first derive the preamble transmission success probability based on the SINR outage, which facilitates our derivations on the RACH success probability by further considering the preamble collision. Note that we just consider intra-cell interference in this work. This is due to the fact that in many real applications, the neighboring BSs in a 4G/5G network use different preambles from different routes and have different RACH times, which makes them non-overlapped. Therefore, it is not common to consider inter-cell interference in the RACH [7]. Thus, the preamble transmission success probability is derived as

$$\mathbb{P}_{S,i}[R_i, n_i] = 1 - \prod_{r_i=1}^{R_i} \Big(1 - \mathbb{P}_i[\theta_{r_i} | N_i = n_i]\Big), \quad (7)$$

where $N_i = n_i$ is the number of intra-cell interfering IoT devices in CE group i (i.e., using the same preamble as the typical IoT device simultaneously in CE group i), and

$$\theta_{r_i} = \Big\{\mathrm{SINR}_{r_i,1} \geq \gamma_{th}, \mathrm{SINR}_{r_i,2} \geq \gamma_{th}, \mathrm{SINR}_{r_i,3} \geq \gamma_{th}, \mathrm{SINR}_{r_i,4} \geq \gamma_{th}\Big\}. \quad (8)$$

In (8), γ_{th} is the SINR threshold, $\mathrm{SINR}_{r_i,1}$, $\mathrm{SINR}_{r_i,2}$, $\mathrm{SINR}_{r_i,3}$, and $\mathrm{SINR}_{r_i,4}$ are the received SINRs of the four continuous symbol groups in the r_ith repetition.

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Based on the Binomial theorem, the preamble transmission success probability in (7) can be rewritten as

$$\mathbb{P}_{S,i}[R_i, n_i] = \sum_{r_i=1}^{R_i} (-1)^{r_i+1} \binom{R_i}{r_i} \mathbb{P}_i[\theta_1, \theta_2, \cdots, \theta_{r_i} | N_i = n_i], \quad (9)$$

where $\mathbb{P}_i[\theta_1, \theta_2, \cdots, \cdots \theta_{r_i} | N_i = n_i]$ is the probability that all $4 \times r_i$ (a preamble consists of 4 preamble symbol groups) preamble symbol groups are successfully transmitted.

Then, we formulate the RACH success probability taking into account both the preamble outage and the collision. The RACH success probability $\mathcal{P}_i^1$ is represented as

$$\mathcal{P}_i^1 = \sum_{n_i=0}^{\infty} \Big\{ \underbrace{\mathbb{P}[N_i = n_i]}_{\text{I}} \underbrace{\mathbb{P}_{S,i}[R_i, n_i]}_{\text{II}} \underbrace{\prod_{j \in \Phi_i} \Big(1 - \mathbb{P}_{S,i}[R_i, n_i]\Big)}_{\text{III}} \Big\}, \quad (10)$$

where part I is the probability that the number of intra-cell interfering IoT devices in CE group i is equal to n_i; part II is the preamble transmission success probability of the typical IoT device conditioning on $N_i = n_i$; and part III is the preamble transmission failure probability that the preambles transmitting from other n_i intra-cell interfering IoT devices in CE group i are not successfully received by the BS.

Considering the approximation of the interference field for each CE group, the Probability Mass Function (PMF) of the number of intra-cell interfering IoT devices in CE group i, i.e., part I in (10) is represented as [14, Eq.(3)]

$$\mathbb{P}[N_i = n_i] = \frac{c^{(c+1)} \Gamma(n_i + c + 1) \Big(\frac{\lambda_i^a}{\lambda_B}\Big)^{n_i}}{\Gamma(c+1)\Gamma(n_i+1) \Big(\frac{\lambda_i^a}{\lambda_B} + c\Big)^{n_i+c+1}}, \quad (11)$$

where λ_i^a is given in (5), $c = 3.575$ is a constant related to the approximate PMF of the PPP Voronoi cell, and $\Gamma(\cdot)$ is the gamma function.

Now, we need to focus on analyzing part II in (10) for three CE groups in the following subsections.

A. CE Group 0, $R_0 \leq 2$ (i = 0)

The SINR received at the typical BS can be written as

$$\text{SINR} = \frac{\rho h_0}{\mathcal{I}_0^{intra} + \sigma^2}, \quad (12)$$

where σ^2 is the noise power at the BS, $\mathcal{I}_0^{intra}$ is the aggregate intra-cell interference of the typical IoT device given as

$$\mathcal{I}_0^{intra} = \sum_{j \in \Phi_0} \rho h_{0,j}. \quad (13)$$

In (13), $h_{0,j}$ is the channel power gain from the jth intra-cell interfering IoT device in CE group 0 to the typical BS.

For ease of presentation, we set $l_0 = 4 \times r_0$, and the preamble transmission success probability is presented in the following **Lemma**.

Lemma 2. *The preamble transmission success probability of a randomly chosen IoT device in CE group 0 is expressed as*

$$\mathbb{P}_{S,0}[R_0, n_0] = \sum_{r_0=1}^{R_0} (-1)^{r_0+1} \binom{R_0}{r_0} \exp\Big(-\frac{l_0 \gamma_{th} \sigma^2}{\rho}\Big) \Big(\frac{1}{1+\gamma_{th}}\Big)^{n_0 l_0}. \quad (14)$$

Proof. See Appendix A. □

Substituting (11) and (14) into (10), we derive the RACH success probability in the following **Theorem**.

Theorem 1. *The RACH success probability of a randomly chosen IoT device in CE group 0 is derived as*

$$\mathcal{P}_0^1 = \sum_{n_0=0}^{\infty} \Bigg\{ \underbrace{\frac{c^{(c+1)} \Gamma(n_0 + c + 1) \Big(\frac{\lambda_0^a}{\lambda_B}\Big)^{n_0}}{\Gamma(c+1)\Gamma(n_0+1) \Big(\frac{\lambda_0^a}{\lambda_B} + c\Big)^{n_0+c+1}}}_{\text{I}} \underbrace{\sum_{r_0=1}^{R_0} (-1)^{r_0+1} \binom{R_0}{r_0} e^{v_0} \Big(\frac{1}{1+\gamma_{th}}\Big)^{n_0 l_0}}_{\text{II}} \underbrace{\Bigg(1 - \sum_{r_0=1}^{R_0} (-1)^{r_0+1} \binom{R_0}{r_0} e^{v_0} \Big(\frac{1}{1+\gamma_{th}}\Big)^{n_0 l_0}\Bigg)^{n_0}}_{\text{III}} \Bigg\}, \quad (15)$$

where $v_0 = -l_0 \gamma_{th} \sigma^2 / \rho$.

B. CE group 1 and CE group 2, $R_i > 2$ (i = 1, 2)

Same as the analysis in Part A, we have

$$\text{SINR}(k) = \frac{P h_{i,k} (x_{i,k})^{-\alpha}}{\mathcal{I}_i^{intra} + \sigma^2}, \quad (16)$$

with $\mathcal{I}_i^{intra}$ given as

$$\mathcal{I}_i^{intra} = \sum_{j \in \Phi_i} P h_{i,j} (x_{i,j})^{-\alpha}. \quad (17)$$

In (17), $h_{i,j}$ and $x_{i,j}$ are channel power gain and distance from the jth intra-cell interfering IoT device in CE group i to the typical BS.

According to the nature of the Poisson Process, given that there are $N_i + 1$ IoT devices in the area of CA_i, $x_{i,k}$ follows independent and identical uniform distribution [15]. Let X_i $(i = 1, 2)$ denote the random variable with the same uniform distribution, the PDF of X_i is derived as

$$f_{X_i}(x_i) = 2x_i / (D_i^2 - D_{i-1}^2), (D_{i-1} \leq x_i \leq D_i), \quad (18)$$

where D_1 is given in (2) and D_2 is given in (3).

We set $l_i = 4 \times r_i$, and derive the preamble transmission success probability in the following **Lemma**.

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Lemma 3. *The preamble transmission success probability of a randomly chosen IoT device in CE group i ($i = 1, 2$) is expressed as*

$$\mathbb{P}_{S,i}[R_i, n_i] = \sum_{r_i=1}^{R_i} (-1)^{r_i+1} \binom{R_i}{r_i} \int_{D_{i-1}}^{D_i} \exp(-l_i v_i \sigma^2) \times \exp(-2\pi \mathcal{A}_i^1 \mathcal{R}_i^1 \lambda_i^a \mathcal{F}) \frac{2x_i}{D_i^2 - D_{i-1}^2} dx_i, \quad (19)$$

where with $\mathcal{F}$ obtained in (B.3) *and* $v_i = \dfrac{\gamma_{th}(x_{i,k})^\alpha}{P}$.

Proof. See Appendix B. □

Substituting (11) and (19) into (10), we derive the RACH success probability in the following **Theorem**.

Theorem 2. *The RACH success probability of a randomly chosen IoT device in CE group i is derived as*

$$\mathcal{P}_i^1 = \sum_{n_i=0}^{\infty} \Bigg\{ \underbrace{\frac{c^{(c+1)}\Gamma(n_i + c + 1)\left(\frac{\lambda_i^a}{\lambda_B}\right)^{n_i}}{\Gamma(c+1)\Gamma(n_i+1)\left(\frac{\lambda_i^a}{\lambda_B} + c\right)^{n_i+c+1}}}_{\text{I}} \underbrace{\sum_{r_i=1}^{R_i} (-1)^{r_i+1} \binom{R_i}{r_i} \int_{D_{i-1}}^{D_i} \Omega_i \frac{2x_i}{D_i^2 - D_{i-1}^2} dx_i}_{\text{II}} \underbrace{\left(1 - \sum_{r_i=1}^{R_i} (-1)^{r_i+1} \binom{R_i}{r_i} \int_{D_{i-1}}^{D_i} \Omega_i \frac{2x_i}{D_i^2 - D_{i-1}^2} dx_i\right)^{n_i}}_{\text{III}} \Bigg\}, \quad (20)$$

where $\Omega_i = \exp(-l_i v_i \sigma^2) \exp(-2\pi \mathcal{A}_i^1 \mathcal{R}_i^1 \lambda_i^a \mathcal{F})$ *with* $v_i = \dfrac{\gamma_{th}(x_{i,k})^\alpha}{P}$ *and* $\mathcal{F}$ *in* (B.3).

IV. Simulation and Discussion

In this section, the derived analytical results are validated via Monte Carlo simulations. The simulation parameters are summarized in Table I following [10]. The BSs and IoT devices are deployed via PPPs in a 10^4 km^2 circle area. We set $\lambda_{\mathrm{B}} = 0.2$ BSs/km^2, $\lambda_{\mathrm{D}} = 10$ IoT devices/km^2, $\gamma_{th} = 33$ dB, $\alpha = 4$, and $\rho = -94$ dBm. The noise is σ^2 = −174+5+10log$_{10}$(180000) = −116.4 dBm and ω = −174+3+10log$_{10}$(3750) = −135.3 dBm. The target SNRs are $\delta_1 = 35.4$ dB and $\delta_2 = 30.4$ dB, respectively.

TABLE I: Simulation Parameters

NB-IoT Bandwidth	180 kHz
NPRACH Subcarrier Spacing	3.75 kHz
Symbol Group	1 CP and 5 symbols
NPRACH Band (3 CE groups)	12 ,12 and 24 subcarriers
Transmission Power	40 dBm (DL), 20 dBm (UL)
Noise Figure	5 dB (DL), 3 dB (UL)

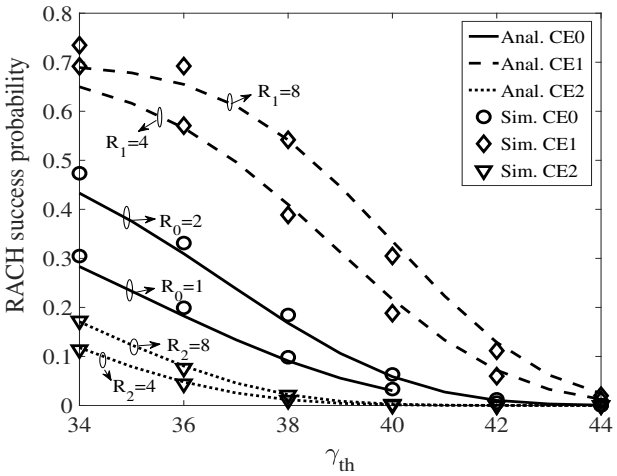


Fig. 3: RACH success probability for three CE groups versus γ_{th}

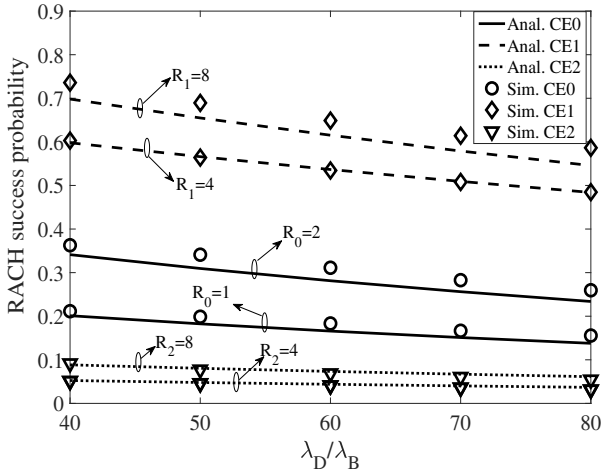


Fig. 4: RACH success probability for three CE groups versus λ_D/λ_B

Fig. 3 plots the RACH success probabilities of a randomly chosen IoT device for three CE groups versus γ_{th}. We first observe that the RACH success probability degrades with the increase of γ_{th}. According to (9), increasing γ_{th} leads to lower preamble transmission success probability of both interfering IoT devices and serving IoT device, thereby decreaseing the overall RACH success probability. Also, we see that the RACH success probabilities of the IoT device in CE group 1 are higher than those in CE group 0, which indicates that increasing the repetition value leads to higher RACH success probability and could ensure the RACH performance with extended coverage.

Fig. 4 plots the RACH success probabilities of a randomly chosen IoT device for three CE groups versus λ_D/λ_B. We first observe that the RACH success probability decreases with the increase of the density ratio between IoT devices and BSs, which is due to the following two reasons: 1) increasing the number of IoT devices generating interference leads to lower received SINR at the BS; 2) increasing the number of IoT devices leads to higher probability of collision. In both

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two figures, increasing repetition value increases the RACH success probability for each CE group, which is due to that it offers more opportunities to re-transmit a preamble with the time and frequency diversity.

V. Conclusion

In this paper, we developed a spatial model to analyze the RACH success probability under the repetition scheme in the NB-IoT network with three CE groups. We first obtained the approximate characterization of interference experienced by a typical IoT device in each CE group. We then derived the analytical expression for the RACH success probability of a randomly chosen IoT device in each CE group taking into account both preamble transmission outage and collision. Our numerical results show that increasing the repetition value leads to higher RACH success probability and could ensure the RACH performance with extended coverage.

Appendix A
A proof of Lemma 2

Using (8) and (12), we have

$$\begin{aligned}&\mathbb{P}_0[\theta_1,\theta_2,\cdots,\theta_{r_0}\,|N_0=n_0]\\ =&e^{\Big(-\frac{l_0\gamma_{th}\sigma^2}{\rho}\Big)}\mathbb{E}\Big[\exp\Big(-\frac{\gamma_{th}}{\rho}\sum_{\beta=1}^{l_0}I_0^{\beta}\Big)\Big|N_0=n_0\Big],\end{aligned}\quad (A.1)$$

where I_0 is the aggregate intra-cell interference in (13). We derive the Laplace transform of the aggregate interference as

$$\begin{aligned}&\mathbb{E}\Big[\exp\Big(-\frac{\gamma_{th}}{\rho}\sum_{\beta=1}^{l_0}I_0^{\beta}\Big)\Big|N_0=n_0\Big]\\ =&\mathbb{E}\Big[\prod_{j=1}^{n_0}\exp\Big(-\gamma_{th}\sum_{\beta=1}^{l_0}h_{0,j}^{\beta}\Big)\Big]\overset{(b)}{=}\Big(\frac{1}{1+\gamma_{th}}\Big)^{n_0l_0},\end{aligned}\quad (A.2)$$

where (b) follows by taking the average with respect to $h_{0,j}^{\beta}$. Substituting (A.1) and (A.2) into (9), the final expression in Lemma 2 is derived.

Appendix B
A proof of Lemma 3

According to (18) and following Appendix A, we have

$$\begin{aligned}p(\gamma_{th})&=\mathbb{E}_{X_i}\Big[\mathbb{P}_{i,k}[\theta_1(k),\theta_2(k),...,\theta_{r_i}(k)|x_i]\Big]\\ &=\int_{D_{i-1}}^{D_i}e^{(-l_iv_i\sigma^2)}\mathbb{E}\Big[e^{\Big(-v_i\sum\limits_{\beta=1}^{l_i}I_{i,k}^{\beta}\Big)}\Big|x_i\Big]f_{X_i}(x_i)dx_i,\end{aligned}\quad (B.1)$$

where $I_{i,k}$ is given in (17) and $v_i=\dfrac{\gamma_{th}(x_{i,k})^{\alpha}}{P}$.

In (B.1), the Laplace Transform of the aggregate intra-cell interference is obtained as

$$\begin{aligned}&\mathbb{E}\Big[e^{\Big(-v_i\sum\limits_{\beta=1}^{l_i}I_{i,k}^{\beta}\Big)}\Big|x_i\Big]\overset{(a)}{=}\mathbb{E}\Big[\prod_{j\in\Phi_i}\Big(\frac{1}{1+v_iP(u_{i,j})^{-\alpha}}\Big)^{l_i}\Big]\\ &\overset{(b)}{=}\exp\big(-2\pi\mathcal{A}_i^1\mathcal{R}_i^1\lambda_i^a\mathcal{F}\big),\end{aligned}\quad (B.2)$$

where (a) is obtained by taking the average with respect to $h_{i,j}^{\beta}$, and (b) follows from the probability generation functional (PGFL) of the PPP obtained as

$$\begin{aligned}\mathcal{F}&=\int_{D_{i-1}}^{D_i}\Big(1-\Big(\frac{1}{1+v_iP(u_i)^{-\alpha}}\Big)^{l_i}\Big)u_idu_i\\ &=\frac{(D_i)^2\Big(-2\Big(\frac{(D_i)^{\alpha}}{Pv_i}\Big)^{l_i}F(D_i)+\alpha l_i+2\Big)}{2\alpha l_i+4}\\ &\quad-\frac{(D_{i-1})^2\Big(-2\Big(\frac{(D_{i-1})^{\alpha}}{Pv_i}\Big)^{l_i}F(D_{i-1})+\alpha l_i+2\Big)}{2\alpha l_i+4},\end{aligned}\quad (B.3)$$

where $F(D_i)={}_2F_1\left(l_i,l_i+\dfrac{2}{\alpha};l+\dfrac{2}{\alpha}+1;-\dfrac{(D_i)^{\alpha}}{Pv_i}\right)$ denotes the Gauss Hypergeometric function. Combing (B.1)–(B.3), we proved Lemma 3 using (9).

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